

Front-End Development

Module : Project Development
Project Title : School Management System

Problem Definition:

The "School Management System" is a web-based application designed to streamline the administration of school activities. It aims to manage student registrations, class schedules, attendance, exams, grades, communication between teachers and parents, and access to educational resources. The system ensures efficient management and quality education.

Requirements:

The School Management System should meet the following requirements:

1. User Registration and Management:

- Allow students, parents, teachers, and administrators to register and create accounts.
- Maintain records of all registered users, including roles and permissions.

2. Class and Schedule Management:

- Create and manage class schedules and timetables.
- Assign teachers to classes and manage class capacities.

3. Attendance Management:

- Track student and teacher attendance.
- Generate attendance reports and notify parents.

4. Exams and Grades Management:

- Schedule exams and manage grading.
- Provide students and parents with access to grades and progress reports.

5. Communication and Notifications:

- Facilitate communication between teachers, parents, and students.
- Send notifications about school events, exams, and other updates.

6. Educational Resources:

- Provide access to learning materials, assignments, and resources.
- Allow teachers to upload and share resources.

7. Fee Management:

- Manage student fee payments and generate invoices.
- Track payment history and send reminders.

8. Event Management:

- Plan and organize school events and activities.
- Manage event registrations and track participation.

9. Data Security and Privacy:

- Ensure secure storage and access of user data.
- Comply with data privacy regulations.

10. Administrator Dashboard:

- Provide real-time metrics on attendance, grades, and school activities.
- Generate reports for decision-making.

User Roles:

1. Administrator:

- Manage student registrations, class schedules, attendance, exams, and grades.
- View and analyze reports on student progress and school activities.
- Handle communication and notifications.

2. Teacher:

- Manage class schedules and student attendance.
- Track student progress, manage exams, and generate reports.
- Communicate with parents and share educational resources.

3. Parent:

- Register and manage their child's account.
- View class schedules, attendance, grades, and progress reports.
- Communicate with teachers and receive notifications.

4. Student:

- Access class schedules, attendance, grades, and educational resources.
- Participate in school events and activities.

Modules and Functionalities:

A. Administrator:

1. Student and Class Management:

- Add, update, and delete student records.
- Create and manage class schedules and timetables.
- Assign teachers to classes.

2. Attendance Management:

- Track student and teacher attendance.
- Generate attendance reports and notify parents.

3. Exams and Grades Management:

- Schedule exams and manage grading.
- Provide access to grades and progress reports.

4. Reports and Analytics:

- Generate reports on attendance, grades, and school activities.
- Analyze data to improve school performance.

5. Communication and Notifications:

- Send notifications about school events, exams, and updates.
- Facilitate communication between teachers, parents, and students.

B. Teacher:

1. Class and Attendance Management:

- Manage class schedules and student attendance.
- Track student progress and manage exams.

2. Educational Resources:

- Upload and share learning materials and assignments.
- Provide feedback to students and parents.

3. Communication:

- Communicate with parents about student progress.
- Send notifications and updates.

C. Parent:**1. Account and Child Management:**

- Register and manage their child's account.
- View class schedules, attendance, grades, and progress reports.

2. Communication:

- Communicate with teachers about their child's progress.
- Receive notifications about school events and updates.

D. Student:**1. Educational Resources:**

- Access learning materials, assignments, and resources.
- Participate in school events and activities.

E. Modules Integration:**1. User Authentication and Access Control:**

- Authenticate users and grant access based on roles.
- Ensure data security and privacy.

2. Student and Class Management:

- Integrate student records with class schedules and attendance tracking.
- Maintain accurate and up-to-date student data.

3. Attendance and Progress Tracking:

- Track student attendance and progress.
- Generate and share reports with parents and administrators.

4. Communication and Notifications:

- Facilitate communication between teachers, parents, and students.
- Send notifications and updates about school activities.

5. Educational Resources:

- Provide access to learning materials and assignments.
- Allow teachers to upload and share resources with students and parents.

6. Reports and Analytics:

- Utilize data from attendance and progress tracking to generate reports.
- Present insights to administrators and teachers for decision-making.